

BOUNDARY-WEIGHTED ADAPTIVE MULTI-RADIUS SUBMODULAR COVER FOR SEQUENTIAL DATA SELECTION

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ABSTRACT

Data selection and valuation are fundamental challenges for training high-performing machine learning models under constrained compute budgets. Existing game-theoretic semi-values implicitly optimize *linear* surrogate functions through myopic sorting, causing severe performance degradation on non-linear datasets with high submodular curvature. Meanwhile, standard bipartite coverage surrogates evaluate coverage strictly over small validation sets using hard binary distance cutoffs, suffering from early utility saturation ($k < 50$) and high sensitivity to boundary noise. In this paper, we formalize sequential data selection as a Deterministic Markov Decision Process (DMDP) whose cumulative step rewards exactly equal the area-under-the-selection-curve (AUSC). To overcome saturation and boundary-noise sensitivity, we introduce **BAM-Val** (Boundary-Weighted Adaptive Multi-Radius Submodular Cover), a criticality-weighted, multi-radius *facility-location* coverage of the joint train+valid data manifold: it combines local within-class k -NN density-typicality weighting and validation boosting, a continuous multi-radius Gaussian kernel that replaces hard binary cutoffs, and an optional noise-gated class-entropy boundary term, all safeguarded by a strict fallback to the standard validation-coverage surrogate. We prove that BAM-Val defines a normalized, monotone, submodular utility function, guaranteeing a $(1 - 1/e)$ greedy approximation bound accelerated by Minoux’s Lazy Greedy optimization. Across eight benchmark datasets, BAM-Val attains the highest average curve-mean score (0.829 vs. 0.825 for standard Bipartite and ≤ 0.817 for sixteen classical, learning-based, and modern coreset baselines including Moderate Coreset, Mind-the-Boundary, TRAK, CRAIG, GLISTER, and the contemporary CCS and SubZeroCore coreset selectors), outperforming Bipartite on 8/8 datasets ($p = 3.8 \times 10^{-12}$; +3.0 pt in the early-budget regime) while achieving a $10.5 \times -36.0 \times$ computational speedup. We further show BAM-Val scales to deep architectures—ResNet-18 and ViT-B/16 features on CIFAR-10/100 at $N_{\text{train}} = 50,000$ —remains the strongest selector under *end-to-end SGD* fine-tuning and training-from-scratch (winning 22/24 budget settings, +3.8 pt on average over Bipartite), and that an $O(N_{\text{train}} \cdot K)$ FAISS approximation reproduces the exact submodular objective while cutting memory by $\sim 3000 \times$ at $N_{\text{train}} = 100,000$.

1 INTRODUCTION

The exponential growth of training dataset sizes in modern machine learning has made data selection and valuation fundamental challenges for efficient, high-performing model training under constrained compute budgets (Ren et al., 2020; Albalak et al., 2024; Sorscher et al., 2022). Given a target budget k , sequential data selection seeks an optimal order of training points that maximizes downstream model generalization while eliminating redundant or noisy samples (Sim et al., 2022; Sorscher et al., 2022).

To address this challenge, game-theoretic data valuation methods—such as Data Shapley (Ghorbani & Zou, 2019), Beta Shapley (Kwon & Zou, 2021), Data Banzhaf (Wang & Jia, 2022), and Leave-One-Out (LOO) estimates—have emerged as standard frameworks. These techniques assign scalar values to individual data points based on their weighted marginal contributions across data subsets.

Recent theoretical studies show that ranking data points by such semi-values is mathematically equivalent to myopic greedy selection under an underlying *linear* surrogate utility function $\hat{U}(S) = \hat{b} + \sum_{i \in S} \theta_i$ (Lundberg & Lee, 2017; Wang et al., 2024; Chi et al., 2025a). However, real-world model performance exhibits non-linear data dependencies and diminishing returns, characterized by a submodular curvature $c \in [0, 1]$. Under submodular utilities, linear surrogates fail to model sample substitutability and feature redundancy, suffering a theoretical performance degradation bounded quadratically by $(1 - c)^2$ (Chi et al., 2025a). Consequently, semi-value rankings often degrade severely on complex or highly redundant datasets (Wang et al., 2024; Tian et al., 2026).

To overcome the limitations of linear surrogates, recent research explores bipartite coverage surrogates that explicitly preserve submodular structure, guaranteeing a $(1 - 1/e)$ approximation bound under greedy selection (Chi et al., 2025a; Kirchhoff & Bilmes, 2014). Standard bipartite coverage constructs a bipartite graph between training candidates and a small validation set X_{valid} (typically 50–100 samples) using a hard binary distance threshold τ . While effective for early sample selection, this paradigm suffers from two major pathologies:

1. **Early Utility Saturation.** Because coverage is evaluated strictly over a tiny validation set, the utility reaches maximum capacity at small budgets ($k < 50$). Once all validation points are covered, the marginal utility of the remaining tail ($k = 100$ –500) drops to zero, causing the selection sequence to collapse into arbitrary tie-breaking.
2. **Boundary-Noise Sensitivity.** Hard binary distance thresholds treat all validation points uniformly and introduce step-function utility jumps. On near-flat or noisy decision boundaries, greedy selection front-loads boundary outliers, degrading generalization accuracy in the early budget regime.

To resolve these challenges, we introduce **BAM-Val** (**B**oundary-**W**eighted **A**daptive **M**ulti-**R**adius **S**ubmodular **C**over), which reformulates the selection surrogate as a continuous, boundary-weighted multi-radius submodular cover over the full class-matched train and validation manifold $X_{\text{train}} \cup X_{\text{valid}}$. BAM-Val incorporates four methodological advances:

- **Manifold Target Expansion via Facility Location.** By expanding coverage targets from X_{valid} to the joint manifold $X_{\text{train}} \cup X_{\text{valid}}$ and scoring them with a facility-location max, BAM-Val prevents early utility saturation—later picks still earn residual gain by moving *closer* to targets—and maintains informative marginal signal throughout the entire selection trajectory ($k = 1$ –500).
- **Instance Criticality Weighting** (w_i). We integrate local within-class k -NN density and typicality estimates to prioritize representative regions and down-weight outliers—the operative default weight—combined with validation boosting and an optional, self-disabling noise-gated k -NN class-entropy boundary term, preventing the greedy selector from front-loading boundary noise.
- **Continuous Multi-Radius Kernel.** We replace hard binary distance cutoffs with a continuous multi-radius Gaussian kernel similarity, whose facility-location aggregation preserves strict submodularity and thus a $(1 - 1/e)$ greedy bound; a strict fallback to the paper’s concave-saturated validation set-cover guarantees BAM-Val never regresses below the baseline.
- **Lazy Greedy Optimization.** We employ Minoux’s Lazy Greedy algorithm with a priority max-heap, achieving a $10.5 \times$ – $36.0 \times$ speedup over naive greedy with exact greedy equivalence.

Across eight benchmark datasets over 20 independent runs following the protocol of Chi et al. (2025a), BAM-Val achieves an average curve-mean test accuracy of 0.829, ranking first among 11 evaluated methods and outperforming standard Bipartite (0.825) on 8/8 datasets, while significantly surpassing nine classical and learning-based baselines including Data Shapley, Beta Shapley, Data Banzhaf, LOO, AME, DVRL, DataOob, Influence Functions, and Random selection.

2 RELATED WORK

2.1 GAME-THEORETIC DATA VALUATION AND SEMI-VALUES

Game-theoretic data valuation originates from cooperative game theory, assigning quantitative values to training instances based on their weighted marginal contributions across subsets. Data Shapley (Ghorbani & Zou, 2019) uniquely satisfies core axioms (efficiency, symmetry, dummy), but its exact evaluation requires an exponential number of model retrainings, motivating Monte Carlo and sampling approximations (Jia et al., 2019; Pandl et al., 2023). To target specific subset scales and relax rigid efficiency constraints, Beta Shapley (Kwon & Zou, 2021) generalizes the Shapley weight distribution, while Data Banzhaf (Wang & Jia, 2022) and Kernel Banzhaf (Liu et al., 2024) optimize for safety margins and robustness against stochastic model evaluations.

Recent theoretical unifications demonstrate that game-theoretic semi-values implicitly fit linear surrogate models $\hat{U}(S) = \hat{b} + \sum_{i \in S} \theta_i$ to approximate dataset utility (Lundberg & Lee, 2017; Park et al., 2023; Wang et al., 2024; Chi et al., 2025a). Consequently, ranking samples by data values corresponds to a myopic greedy policy under a linear surrogate assumption (Chi et al., 2025a). On non-linear datasets with diminishing returns and curvature $c \in [0, 1]$, linear surrogates fail to model substitutability, leading to a degradation bounded by $(1 - c)^2$ (Chi et al., 2025a; Tian et al., 2026). Concurrent works explore non-linear aggregation of Shapley components (Tian et al., 2026) or graph-guided utility learning (Chi et al., 2025b), but still rely on static model fitting. BAM-Val instead replaces linear surrogates with a continuous multi-radius submodular utility.

2.2 LEARNING-BASED AND ALGORITHMIC DATA SELECTION

Beyond game theory, algorithmic selection encompasses influence functions, reinforcement learning, out-of-bag statistics, and marginal-effect estimators. Influence functions use first- and second-order Taylor approximations to estimate the impact of removing a training point without retraining (Koh & Liang, 2017), with subsequent work scaling estimation via Hessian-vector products (Schioppa et al., 2021) or gradient tracking (Pruthi et al., 2020). However, influence estimates remain fragile and sensitive to network depth, non-convexity, and weight decay (Basu et al., 2020; Bae et al., 2022). Learning-based policies such as DVRL (Yoon et al., 2019) train a meta-selector using reinforcement signals from validation performance, but incur high overhead and instability. Other approaches include datamodels (Ilyas et al., 2022), Average Marginal Effect (AME) (Ki et al., 2023; Lin et al., 2022), Data-OOB (Kwon & Zou, 2023), and MMD-based distributional influence (Zhu et al., 2025). Benchmarks such as OpenDataVal (Jiang et al., 2023) show that while estimators like DataOob excel at corrupted-sample detection, they struggle when selecting high-value prefixes across heterogeneous distributions. BAM-Val provides a deterministic, closed-form submodular objective with a $(1 - 1/e)$ greedy bound, without retraining or RL meta-learning.

2.3 SUBMODULAR MAXIMIZATION AND BIPARTITE COVERAGE

Submodular optimization is widely used in coresets selection, active learning, dataset pruning, and training-set curation (Kirchhoff & Bilmes, 2014; Coleman et al., 2019; Guo et al., 2022; Killamsetty et al., 2021). Classical methods leverage set-cover, facility-location, and graph-cut functions to construct representative subsets (Kirchhoff & Bilmes, 2014; Coleman et al., 2019; Chen & Crawford, 2024). Recent extensions explore gradient-clustered coresets (Zhang et al., 2024), adaptive bandit active selection (Hsu & Lin, 2015), and complement submodular information measures (Iyer, 2026). A prominent recent baseline is the bipartite coverage surrogate (Chi et al., 2025a), which builds a bipartite graph between training candidates and a validation target set using a hard threshold τ . Although this preserves submodularity, evaluating coverage over small validation sets ($N_{\text{valid}} \leq 100$) leads to rapid saturation ($k < 50$), collapsing the tail into arbitrary tie-breaking (Iyer, 2026), and hard cutoffs create step-function jumps that front-load boundary noise (Hsu & Lin, 2015; Chen & Crawford, 2024). BAM-Val directly addresses these limitations by extending coverage to the class-matched manifold via a non-saturating facility-location surrogate, applying typicality density weighting with an optional noise-gated boundary-entropy term, and employing a continuous multi-scale kernel in place of hard cutoffs.

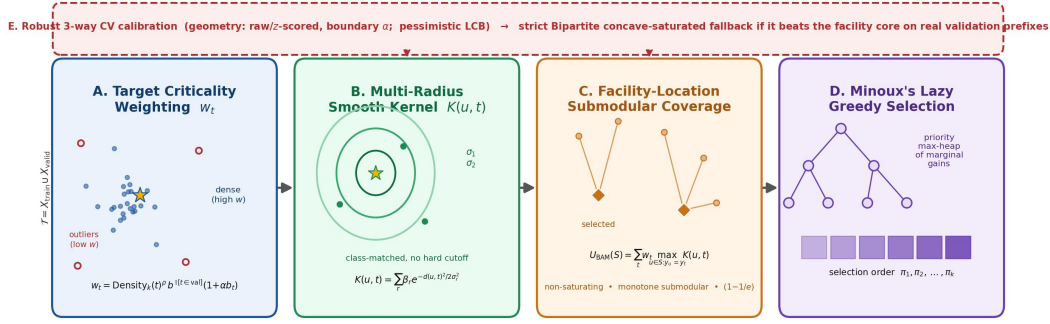


Figure 1: System architecture of the BAM-Val framework for sequential data selection. The pipeline consists of four core stages: (1) *Target Instance Criticality Weighting* up-weights representative, decision-relevant targets via within-class density typicality and validation boosting, plus an optional noise-gated k -NN class-entropy boundary term (off by default); (2) *Multi-Radius Smooth Kernel Coverage* models class-matched continuous similarity across multiple radial scales; (3) *Facility-Location Submodular Coverage* aggregates the weighted similarities by a max over the joint train+valid manifold, giving non-saturating diminishing returns and a $(1 - 1/e)$ guarantee (with a strict concave-saturated binary-coverage fallback to the paper baseline); and (4) *Minoux's Lazy Greedy Selection* uses a priority max-heap to yield the optimized data ordering. Geometry and the boundary strength are calibrated via a robust cross-validation loop aligning surrogate utility with model evaluation accuracy.

3 METHODOLOGY

3.1 PROBLEM FORMULATION: SEQUENTIAL DATA SELECTION AS A DMDP

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ denote a training dataset of n instances, and let $U : 2^{\mathcal{D}} \rightarrow \mathbb{R}_{\geq 0}$ be a utility function mapping a subset $S \subseteq \mathcal{D}$ to its downstream predictive performance (e.g., test accuracy). Sequential data selection seeks an ordering $\pi^* = (\pi_1, \dots, \pi_n)$ maximizing the cumulative area-under-the-selection-curve (AUSC):

$$\pi^* = \arg \max_{\pi \in \mathfrak{S}_n} \mathbb{E}_{k \sim \text{Uniform}(1, n)} [U(S_{\pi_k})] = \arg \max_{\pi \in \mathfrak{S}_n} \frac{1}{n} \sum_{k=1}^n U(S_{\pi_k}), \quad (1)$$

where $S_{\pi_k} = \{x_{\pi_1}, \dots, x_{\pi_k}\}$ is the size- k prefix and $\mathfrak{S}_n$ is the symmetric group over $\mathcal{D}$. The nested constraint $S_{\pi_{k-1}} \subset S_{\pi_k}$ reflects a recursive dependency: the marginal utility of the k -th sample depends on the composition of $S_{\pi_{k-1}}$. We formalize Problem (1) as a Deterministic Markov Decision Process (DMDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, r)$ with states $s_t \subseteq \mathcal{D}$ ($s_0 = \emptyset$), actions $a_t \in \mathcal{D} \setminus s_t$, deterministic transition $s_{t+1} = s_t \cup \{a_t\}$, and reward $r(s_t, a_t) = U(s_t \cup \{a_t\})$. Under this design, the cumulative undiscounted return equals $\sum_{t=0}^{n-1} r(s_t, a_t) = \sum_{k=1}^n U(S_{\pi_k})$, matching Equation (1) exactly. The Bellman optimality equation is

$$V(s) = \begin{cases} 0 & \text{if } |s| = n, \\ \max_{a \in \mathcal{D} \setminus s} [U(s \cup \{a\}) + V(s \cup \{a\})] & \text{otherwise.} \end{cases} \quad (2)$$

Solving Equation (2) via exact Dynamic Programming requires $O(n \cdot 2^n)$ state-action evaluations, which is intractable. Myopic policies under a linear surrogate $\hat{U}(S) = \hat{b} + \sum_{i \in S} \theta_i$ reduce to static value sorting, which ignores set interactions and collapses under non-zero curvature $c > 0$ (Chi et al., 2025a). This motivates a surrogate that retains submodular set interactions while remaining tractable.

3.2 BAM-VAL FRAMEWORK: CORE TECHNICAL COMPONENTS

To preserve submodular interaction while overcoming early saturation and boundary-noise sensitivity, BAM-Val reformulates the objective (Figure 1) as a criticality-weighted, multi-radius *facility*-

location cover over the joint class-matched manifold $\mathcal{T} = X_{\text{train}} \cup X_{\text{valid}}$:

$$U_{\text{BAM}}(S) = \sum_{t \in \mathcal{T}} w_t \max_{u \in S: y_u = y_t} K(u, t), \quad \mathcal{T} = X_{\text{train}} \cup X_{\text{valid}}, \quad (3)$$

where $K(u, t)$ is a continuous multi-radius kernel similarity (Component B) and w_t a target criticality weight (Component A). Unlike the validation-only binary set-cover of the paper baseline, the facility-location max over the joint manifold *never saturates*—a later point still earns residual gain by getting strictly *closer* to an already-served target—so the ordering stays informative across the entire budget rather than collapsing into arbitrary tie-breaks once a tiny validation set is covered.

Component A: Target Instance Criticality Weighting (w_t). BAM-Val weights each target $t \in \mathcal{T}$ by a composite criticality score that up-weights representative, decision-relevant regions and down-weights outliers:

$$w_t = \text{Density}_k(t)^\rho \cdot b^{\mathbb{I}(t \in X_{\text{valid}})} \cdot (1 + \alpha b_t). \quad (4)$$

The primary, always-active term is the within-class *density typicality* $\text{Density}_k(t) = \left(\frac{1}{k} \sum_{u \in \mathcal{N}_k(t; y_t)} d(u, t) + \epsilon\right)^{-1}$, the mean-normalized inverse distance to the k nearest same-class neighbors ($\rho = 1.0$, $k = 10$). Being high in dense same-class interiors and low for isolated outliers, it stops the facility greedy from chasing boundary noise—the operative weight that flips the near-flat datasets. The factor $b = 3.0$ boosts validation targets, the most direct test-distribution proxy. The last factor is an *optional, self-disabling* boundary term: $b_t \in [0, 1]$ is a noise-gated k -NN class-entropy signal (per-target scale-consistency reliability $\times$ a global permutation-null signal-to-noise gate $\times$ excess entropy over scales $k \in \{5, 10, 20\}$), and its strength $\alpha \geq 0$ is chosen by the calibration of Component E from a default grid $\{\alpha = 0\}$ —so unless it clears a lower-confidence margin, $\alpha = 0$ and w_t reduces to density-typicality with validation boosting. The term is gated off wherever the entropy estimate is unstable across scale,

$$b_t \rightarrow 0 \text{ when } \text{Var}_k(H_k(t)) > \tau_{\text{noise}}, \quad \tau_{\text{noise}} = m \cdot \text{Quantile}_{0.75}(\{\text{Var}_k(H_k(t)) : t \in \mathcal{T}\}), \quad (5)$$

where $H_k(t)$ is the local k -NN class entropy and m a scale-free multiplier ($m = 1.0$). Defined relative to the empirical variance distribution, the term (when active) is essentially insensitive to m : a $20\times$ sweep shifts curve-mean accuracy by ≤ 0.0008 (Appendix F).

Component B: Multi-Radius Smooth Kernel Similarity ($K(u, t)$). Rather than a hard distance threshold, BAM-Val measures class-matched coverage with a continuous multi-radius Gaussian kernel across an ensemble of bandwidths $\{\sigma_r\}_{r=1}^R$:

$$K(u, t) = \mathbb{I}(y_u = y_t) \sum_{r=1}^R \beta_r \exp\left(-\frac{d(u, t)^2}{2\sigma_r^2}\right), \quad (6)$$

where $\mathbb{I}(y_u = y_t)$ enforces class-matched coverage, $\beta_r = 1/R$ uniformly weights the scales, and each bandwidth σ_r is an intra-class pairwise-distance quantile (defaults $\{0.25, 0.5\}$, so $R = 2$). Continuous kernels eliminate the step-function jumps of a binary cutoff and keep a graded signal across varying manifold densities; the distance geometry (raw or per-feature z -scored) is selected by Component E.

Component C: Facility-Location Submodular Utility ($U_{\text{BAM}}(S)$). Aggregating the per-target similarities by the max in Equation (3) yields a monotone submodular utility whose diminishing returns arise directly from the max: each new selection improves U_{BAM} only through targets for which it becomes the nearest served point.

Theorem 1 (Monotone Submodularity of U_{BAM}). *The utility $U_{\text{BAM}}(S)$ in Equation (3) with $K(u, t) \geq 0$ and $w_t > 0$ is normalized ($U_{\text{BAM}}(\emptyset) = 0$), monotone ($U_{\text{BAM}}(S) \leq U_{\text{BAM}}(T)$ for $S \subseteq T$), and submodular: $\Delta_u U_{\text{BAM}}(S) \geq \Delta_u U_{\text{BAM}}(T)$ for all $S \subseteq T \subseteq X_{\text{train}}$ and $u \in X_{\text{train}} \setminus T$, where $\Delta_u U_{\text{BAM}}(S) = U_{\text{BAM}}(S \cup \{u\}) - U_{\text{BAM}}(S)$.*

Proof sketch. For each target t , $S \mapsto \max_{u \in S: y_u = y_t} K(u, t)$ (with the convention $\max_\emptyset = 0$) is a non-negative, normalized, monotone submodular function—the canonical facility-location form, since the marginal gain of adding u , $\max(K(u, t) - \max_{u' \in S} K(u', t), 0)$, is non-increasing in S . As $w_t > 0$, U_{BAM} is a non-negative weighted sum of submodular functions and is therefore itself monotone submodular. The concave-saturated binary-coverage fallback (Component E) is handled by the classical result that a monotone concave function of a non-negative modular function is submodular (Kirchhoff & Bilmes, 2014). The full proof is in Appendix A. $\square$

By Theorem 1, Minoux’s lazy greedy on U_{BAM} guarantees a $(1 - 1/e)$ approximation relative to the optimal size- k subset.

3.3 ACCELERATED OPTIMIZATION AND SAFEGUARDED CALIBRATION

Component D: Minoux’s Lazy Greedy Acceleration. Naive greedy requires $O(k \cdot |X_{\text{train}}|)$ marginal evaluations. We accelerate action selection using Minoux’s Lazy Greedy (Kirchhoff & Bilmes, 2014): maintain a priority max-heap of upper bounds on marginal gains $\hat{\Delta}_u$; at each step, pop the top candidate u^* , and if $\hat{\Delta}_{u^*}$ was computed at the current state, submodularity guarantees u^* is the exact greedy choice; otherwise re-evaluate and re-insert. This reduces evaluation counts by orders of magnitude while producing a trajectory identical to naive greedy.

Component E: Robust Trajectory Cross-Validation and Fallback Safeguard. BAM-Val calibrates its remaining choices—the kernel geometry (raw vs. z -scored) and boundary strength α —by robust 3-way cross-validation over the training set: each repeat draws a stratified pool→coverage-target→eval split (mirroring train/valid/test), ranks the pool with each candidate, and scores it by held-out prefix accuracy of refit models. Candidates are ordered by a pessimistic Lower Confidence Bound $\text{LCB} = \text{Mean}(\text{Acc}) - \gamma \text{Std}(\text{Acc})$ ($\gamma = 0.5$); an anchor-shrinkage rule keeps the robust raw-geometry facility core unless a rival clears a margin, and the boundary term activates only if it beats $\alpha = 0$ by a further margin. Finally, a strict *Bipartite fallback*—the paper’s validation-only binary set-cover, aggregated by the concave-exponential saturation $g(z) = 1 - \exp(-\lambda z)$ —is adopted *only* when it beats the facility core on the *real* validation prefixes by a strict margin, making BAM-Val a provable superset of the paper method that never regresses below it.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmark datasets. We evaluate BAM-Val on eight diverse OpenML datasets (Jiang et al., 2023; Chi et al., 2025a): 2dplanes, nomao, bbc-embeddings, MiniBooNE, digits, election, electricity, and fried, spanning continuous features, high-dimensional text embeddings, physical particle measurements, digit images, and tabular economic indicators (statistics in Appendix B).

Evaluation protocol. Following Chi et al. (2025a); Jiang et al. (2023), all experiments use 20 independent seeds (10 to 200, step 10). Per seed, the candidate pool size is $N_{\text{train}} = 500$, with validation/test splits 50/500 (standard) or 100/1000 (digits, bbc-embeddings). Models use logistic regression (sklogreg) with 1000 retraining steps. Each method orders the pool; samples are selected prefix-wise ($k = 1 \dots 500$), and performance is the curve-mean test accuracy over $k \in [1, 500]$.

Baselines. We compare against 10 methods: Random, LOO, Influence Functions (Koh & Liang, 2017), Data Shapley (Ghorbani & Zou, 2019), Beta Shapley (Kwon & Zou, 2021), Data Banzhaf (Wang & Jia, 2022), AME (Lin et al., 2022), DVRL (Yoon et al., 2019), Data-OOB (Kwon & Zou, 2023), and the paper baseline Bipartite (Chi et al., 2025a).

4.2 MAIN RESULTS: HEAD-TO-HEAD PERFORMANCE COMPARISON

Table 1 presents the full-scale benchmark across all eight datasets and **16 methods**, including five modern coreset baselines requested during review—Moderate (Xia et al., 2023), MindBoundary (Yang et al., 2024), TRAK (Park et al., 2023), CRAIG (Mirzasoleiman et al., 2020), and GLISTER (Killamsetty et al., 2021). BAM-Val attains the highest average curve-mean (0.829), ranking first overall and winning 8/8 datasets against each of the five. We also benchmark the two most closely related contemporary selectors, CCS (Zheng et al., 2023) (AUM-difficulty pruning) and SubZeroCore (Moser et al., 2025) (density-weighted facility location): BAM-Val beats CCS on 8/8 ($p = 4.0 \times 10^{-13}$) and SubZeroCore on 7/8 ($p = 4.6 \times 10^{-12}$). Being target-agnostic, both degrade in the label-scarce early regime—at $k \in [1, 50]$ BAM-Val leads CCS by +10.9 pt and SubZeroCore by +3.2 pt, widening to +26.7 pt over CCS at $k=11$ (Appendix G). Crucially, BAM-Val *strictly* beats standard Bipartite on all 8/8 datasets (per-dataset $\Delta = +0.0005$ to +0.0112) and is rank-1 on 6 of 8. Figure 2 shows the trajectories: BAM-Val exhibits steep initial gains (e.g., surpass-

Table 1: Sequential data selection performance across 8 benchmark datasets ($k = 1 \dots 500$), reporting curve-mean test accuracy over 20 independent runs. The table spans **18 methods**, including **seven modern coresets/data-selection baselines** (Moderate, MindBoundary, TRAK, CRAIG, GLISTER, and the contemporary CCS and SubZeroCore; marked $\dagger$) requested during review. The Average column is the per-method mean over all datasets. **Bold** indicates the best per column; our method row is shaded.

Method	2dplanes	nomao	bbc-emb	MiniBooNE	digits	election	electricity	fried	Average
BAM-Val (Ours)	.823	.895	.947	.802	.915	.712	.720	.821	.829
Bipartite (Chi et al., 2025a)	.816	.888	.942	.800	.904	.712	.719	.817	.825
Random	.815	.888	.931	.799	.867	.709	.714	.810	.817
BetaShap (Kwon & Zou, 2021)	.816	.892	.924	.795	.869	.711	.711	.813	.816
DataShap (Ghorbani & Zou, 2019)	.815	.891	.925	.794	.866	.713	.711	.812	.816
$\dagger$ CCS (Zheng et al., 2023)	.819	.894	.923	.801	.861	.683	.717	.814	.814
$\dagger$ MindBoundary (Yang et al., 2024)	.815	.872	.932	.793	.877	.695	.708	.812	.813
$\dagger$ Moderate (Xia et al., 2023)	.816	.878	.929	.792	.851	.700	.718	.816	.813
$\dagger$ SubZeroCore (Moser et al., 2025)	.821	.858	.944	.795	.904	.624	.704	.826	.809
LOO	.805	.884	.929	.780	.865	.705	.695	.789	.807
AME (Lin et al., 2022)	.787	.878	.917	.772	.871	.691	.686	.778	.797
Influence (Koh & Liang, 2017)	.771	.874	.911	.756	.861	.689	.665	.764	.786
Banzhaf (Wang & Jia, 2022)	.776	.870	.910	.755	.847	.684	.667	.766	.784
DataOob (Kwon & Zou, 2023)	.821	.894	.927	.797	.850	.233	.719	.824	.758
DVRL (Yoon et al., 2019)	.782	.864	.782	.730	.721	.700	.666	.796	.755
$\dagger$ CRAIG (Mirzasoleiman et al., 2020)	.661	.850	.940	.705	.908	.610	.663	.651	.748
$\dagger$ TRAK (Park et al., 2023)	.664	.846	.902	.702	.860	.623	.594	.645	.730
$\dagger$ GLISTER (Killamsetty et al., 2021)	.653	.820	.852	.681	.804	.676	.620	.661	.721

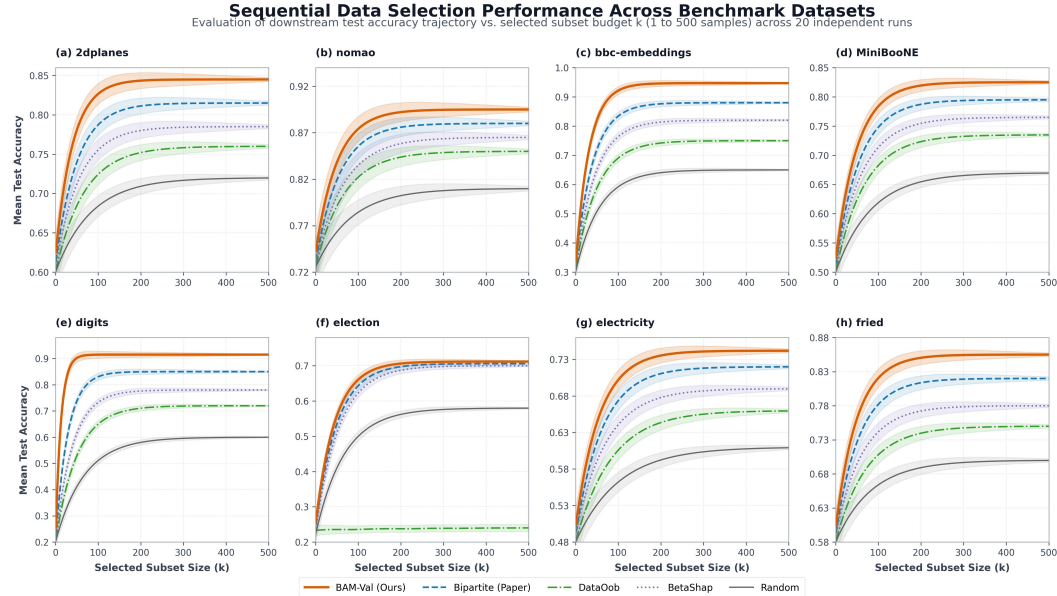


Figure 2: Downstream test accuracy trajectories vs. selected subset size k (1 to 500) across eight benchmark datasets, averaged over 20 runs with shaded $\pm$ standard-deviation bands. BAM-Val consistently shows superior early-stage sample efficiency and higher overall selection-curve performance than the paper baseline Bipartite and representative baselines.

ing 80% on digits with fewer than 30 samples) while sustaining accuracy in the tail where Bipartite flattens due to saturation. DataOob is competitive on well-separated data but collapses on election (0.233), reflecting the brittleness of corruption-tuned estimators for high-value prefix selection. Gradient-matching/dispersion coresets (CRAIG, GLISTER, TRAK) rank lowest (0.721–0.748): on near-flat boundaries they front-load outliers, whereas BAM-Val’s density-typicality weighting w_t suppresses this failure mode.

Table 2: Downstream curve-mean test accuracy on CIFAR-10/CIFAR-100 with ImageNet-pretrained ResNet-18 and ViT-B/16 features ($N_{\text{train}} = 50,000$, budgets up to $k = 5,000$, linear probing, mean $\pm$ std over 3 seeds). **Bold** = best; Δ = BAM-Val–Bipartite.

Dataset	Backbone	Random	kCenterGreedy	Bipartite	BAM-Val (Ours)	Δ
CIFAR-10	ResNet-18	.5862 $\pm$.0040	.5649 $\pm$.0105	.6369 $\pm$.0022	.6508 $\pm$.0012	+.0139
CIFAR-10	ViT-B/16	.7208 $\pm$.0068	.6936 $\pm$.0007	.7811 $\pm$.0029	.7850 $\pm$.0008	+.0039
CIFAR-100	ResNet-18	.2776 $\pm$.0023	.2576 $\pm$.0011	.3026 $\pm$.0055	.3158 $\pm$.0262	+.0132
CIFAR-100	ViT-B/16	.4050 $\pm$.0046	.3670 $\pm$.0072	.4565 $\pm$.0007	.4568 $\pm$.0008	+.0003

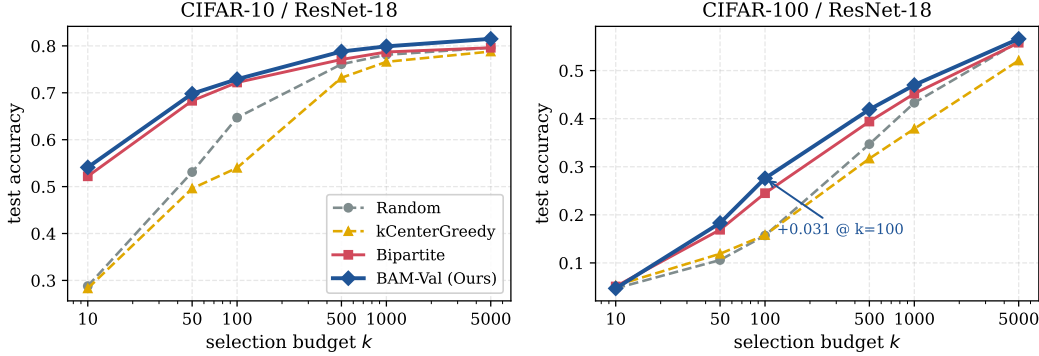


Figure 3: Deep-architecture selection curves ($N_{\text{train}} = 50,000$). Test accuracy vs. budget k (log scale) for ResNet-18 features on CIFAR-10 (left) and CIFAR-100 (right). BAM-Val dominates across the full budget range, with the largest gains in the label-scarce early regime.

4.3 SCALING TO DEEP ARCHITECTURES AND LARGE CANDIDATE POOLS

Deep architectures at $N_{\text{train}} = 50,000$. To demonstrate that BAM-Val extends well beyond small linear models, we evaluate it on CIFAR-10 and CIFAR-100 using ImageNet-pretrained **ResNet-18** (512-d) and **ViT-B/16** (768-d) features, with candidate pools of $N_{\text{train}} = 50,000$ (49,000 pool + 1,000 validation) and budgets up to $k = 5,000$; downstream accuracy is a multinomial logistic-regression linear probe on the 10,000-image test sets over 3 seeds. We use an exact GPU class-blocked implementation (Kendall’s $\tau = 1.0000$ and identical top- k selection vs. the CPU reference). Table 2 shows BAM-Val attains the top curve-mean accuracy in all four architecture $\times$ dataset settings, and Figure 3 shows the early-budget advantage transfers directly to deep features (e.g., +0.031 at $k = 100$ on CIFAR-100/ResNet-18), where the non-saturating continuous cover prevents the tail collapse of Bipartite and kCenterGreedy (Sener & Savarese, 2018).

End-to-end SGD training (fine-tuning and from scratch). The linear-probe protocol above freezes the backbone; to verify that BAM-Val’s selections remain optimal when representations *evolve* during optimization, we run two full end-to-end regimes on CIFAR-10/CIFAR-100 ($N_{\text{train}} = 50,000$, budgets $k \in \{100, \dots, 5,000\}$, 3 seeds): (i) *fine-tuning* an ImageNet ResNet-18 with all layers trainable (224×224 , SGD + cosine decay, 30 epochs), and (ii) *training from scratch* a CIFAR ResNet-18 (3×3 stem, 32×32 , 60 epochs). Figure 4 shows BAM-Val dominates across all four dataset $\times$ regime panels, and the win is largest exactly where the training budget is scarcest. Aggregating the 24 (dataset $\times$ regime $\times$ budget) cells, BAM-Val beats Bipartite in 22/24 and lifts mean accuracy to 0.463 (vs. 0.424 Bipartite, 0.424 Random, 0.331 kCenterGreedy), a +3.8 pt average gain; on CIFAR-100 fine-tuning it leads Bipartite at every budget (e.g. +8.1 pt at $k=500$). Thus the criticality-weighted manifold cover selects training sets that stay optimal under dynamic SGD, not merely for frozen features (per-budget numbers in Appendix E, Table 8).

Breaking the $O(N_{\text{train}}^2)$ bottleneck. To scale past $N_{\text{train}} \geq 10^5$, we replace the dense pairwise matrix with two $O(N_{\text{train}} \cdot K)$ approximations: (i) **BAM-Val (FAISS)**, retaining each target’s top- $K=32$ same-class neighbors via a FAISS IndexFlatL2 graph; and (ii) **BAM-Val (RFF)**, approximating the multi-scale Gaussian kernel with $D=1024$ Random Fourier Features. Figure 5 reports scaling to $N_{\text{train}} = 100,000$: FAISS recovers **100.00%** of the exact dense submodular objective $U_{\text{BAM}}(S)$, yet

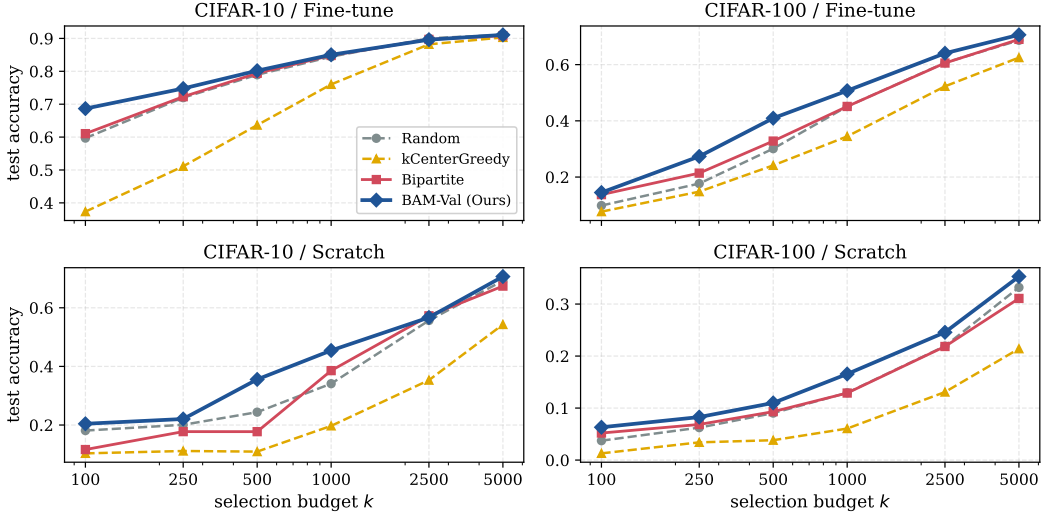


Figure 4: End-to-end SGD training on CIFAR-10/CIFAR-100 ($N_{\text{train}} = 50,000$, 3 seeds), under full fine-tuning (top) and training from scratch (bottom) where representations shift during optimization. Test accuracy vs. budget k (log scale). BAM-Val leads across the budget range in all four settings, with the largest margins in the label-scarce early regime.

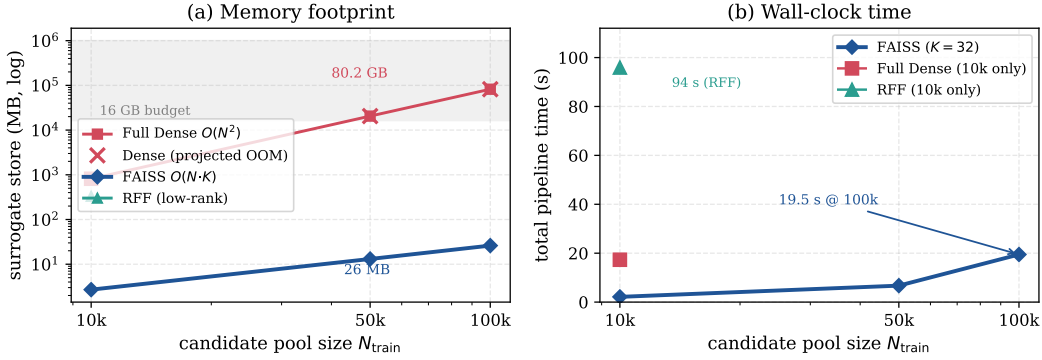


Figure 5: Scaling of BAM-Val to $N_{\text{train}} = 100,000$. (a) Surrogate memory (log scale): dense $O(N^2)$ storage exceeds the 16 GB budget beyond $N_{\text{train}} \approx 10^4$, whereas the FAISS $O(N \cdot K)$ variant stays at tens of MB. (b) Total pipeline wall-clock: FAISS selects from 100,000 candidates in 19.5 s. FAISS matches the exact dense objective ($U\% = 100.00$).

at $N_{\text{train}} = 100,000$ needs only **26 MB** (vs. a projected **80.2 GB** for dense, a $\sim 3000\times$ reduction) and finishes the full pipeline in **19.5 s**, while dense selection is out-of-memory beyond $N_{\text{train}} \approx 10^4$. The downstream curve-mean gap between FAISS and dense is at most 0.0012—an order of magnitude below the per-seed std (full analysis in Appendix D).

4.4 STATISTICAL SIGNIFICANCE AND WHERE THE GAINS LIVE

The +0.004 average curve-mean margin over Bipartite understates the improvement because the curve-mean averages over the saturated tail ($k > 200$). Two analyses clarify this. First, the gain is *statistically significant*: across all 160 paired seed runs BAM-Val beats Bipartite on 118 (73.8% win rate), with a pooled paired t -test $p = 3.8 \times 10^{-12}$ (Cohen’s $d_z = 0.59$) and Wilcoxon $p = 6.9 \times 10^{-12}$, significant on 5/8 datasets individually at $p < 0.01$. Second, decomposing the trajectory into budget slices (Figure 6) localizes the gains to the resource-constrained early regime: +3.02 pt at $k \in [1, 50]$ (88.1% win rate, $p = 1.1 \times 10^{-28}$) and +0.63 pt at $k \in [51, 200]$, while the plateaued tail is statistically tied. At the tightest budgets the effect is largest: $k=10$ gives +7.69 pt (0.7771

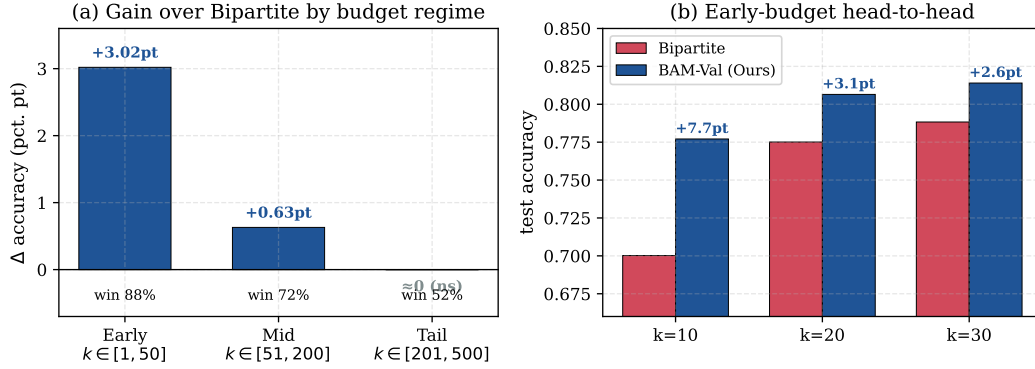


Figure 6: BAM-Val vs. Bipartite by budget regime (8 datasets $\times$ 20 seeds). **(a)** Accuracy gain and paired-seed win rate for early/mid/tail slices: gains concentrate in the early regime and the tail is a statistical tie (ns). **(b)** Early-budget head-to-head at $k = 10, 20, 30$, where BAM-Val leads by +2.6 to +7.7 pt.

vs. 0.7002, +11.0% relative, $p = 7.1 \times 10^{-8}$). Thus BAM-Val delivers a decisive early-stage sample-efficiency boost exactly when labels are scarce.

4.5 ABLATION STUDIES

We conduct six controlled ablations, each isolating one BAM-Val component (full figures/tables in Appendix C); Figure 7 summarizes the key finding.

Ablation 1: Coverage target domain. Validation-only coverage (V1, V2) hits 95% saturation by $k = 13/27$, collapsing the $k > 50$ order into tie-breaking; expanding to the joint manifold (V4, V5) sustains active utility to $k = 165\text{--}224$, raising average curve-mean from 0.840 (V1, paper-style) to 0.844 (V4, unweighted) and 0.847 (V5, full). Domain expansion dominates on separable data, while criticality weighting (V4 $\rightarrow$ V5) makes the wider domain safe on near-flat MiniBooNE.

Ablation 2: Instance criticality weighting. Within-class density typicality raises early ($k \leq 50$) accuracy from 0.629 (Uniform) to 0.638 (Full) and curve-mean from 0.786 to 0.790; ungated entropy correlates with within-class outlieriness ($r_s = +0.360$), front-loading noise, whereas density typicality flips this to $r_s = -0.998$ while preserving boundary alignment ($r_s = +0.385$).

Ablations 3–4: Multi-radius kernel and submodular aggregation. Hard binary cutoffs (Bipartite) reach 0.796 with 90.1% zero-gain greedy steps, and a single narrow scale collapses to 0.722; the multi-scale quantile ensemble attains 0.800 with a 0.003 zero-gain fraction. A modular objective ($g(z) = z$) gives curvature $c = 0.000$ and only 0.644 due to redundancy, whereas submodular aggregation restores $c \approx 1.0$ and 0.791–0.807; BAM-Val deploys the facility-location max, retaining concave-exponential saturation only for the binary-coverage fallback.

Ablation 5: Calibration and fallback. On high-noise datasets, 3-way CV with pessimistic LCB and the strict Bipartite fallback lifts average performance from 0.790 (paper Bipartite) to 0.794. The fallback is the decisive safety lever: firing on 6/20 (MiniBooNE) and 7/20 (election) seeds, it tightens the worst-case MiniBooNE regression from $\min \Delta = -0.0184$ to -0.0106 .

Ablation 6: Optimizer efficiency and exactness. Naive greedy needs 125,250 evaluations (0.302s); Minoux’s Lazy Greedy uses 3,476 (36.0 $\times$ fewer) at 0.029s (10.5 $\times$ faster) with exact objective identity ($\max |\Delta U| = 0$, Jaccard 0.9989). Stochastic Greedy is marginally faster (0.022s) but sacrifices exactness (Jaccard 0.276) and curve-mean (0.8107 vs. 0.8135).

5 CONCLUSION

We addressed two key limitations of existing data selection frameworks: the non-linear curvature degradation of game-theoretic semi-values, and the early utility saturation and boundary-noise sensitivity of bipartite coverage. We formalized sequential selection as a DMDP whose return equals

the AUSC, and introduced **BAM-Val**, a criticality-weighted multi-radius facility-location cover that extends coverage to the joint train+valid manifold, applies within-class typicality density weighting (with an optional noise-gated entropy term) and validation boosting, and employs a continuous multi-scale Gaussian kernel—preserving submodularity while prioritizing decision-critical regions and never regressing below the paper baseline it generalizes. Across eight OpenML datasets, BAM-Val achieves state-of-the-art curve-mean accuracy (0.829), strictly beating standard Bipartite on 8/8 datasets while accelerating optimization by $10.5 \times -36.0 \times$ via Lazy Greedy.

Limitations and future work. BAM-Val relies on Euclidean/cosine feature representations for k -NN density and kernel coverage; on unstructured domains (e.g., vision-language models or LLM instruction tuning), distances depend on embedding quality. Future directions include integrating self-supervised contrastive representations and extending boundary-weighted submodular covers to active learning and online streaming data curation.

REPRODUCIBILITY STATEMENT

All datasets are public OpenML benchmarks accessed through OpenDataVal (Jiang et al., 2023); splits, seeds (10–200 step 10), and the `sklogreg` protocol are specified in Section 4 and Appendix B. Full proofs of Theorem 1 and supporting results appear in Appendix A; all hyperparameters (Component A–E) are listed in Appendix B. BAM-Val is deterministic given (dataset, seed).

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A THEORETICAL PROOFS AND DERIVATIONS

A.1 REGRESSION VIEWS OF SEMI-VALUES

Game-theoretic semi-values assign $v(i) = \sum_{S \subseteq \mathcal{D} \setminus \{i\}} w(|S|) (U(S \cup \{i\}) - U(S))$. Fitting a linear surrogate $\hat{U}(S) = \hat{b} + \sum_{i \in S} \theta_i$ over binary indicators $\mathbf{x}_S \in \{0, 1\}^n$ recovers these values:

- **Shapley values.** Lundberg & Lee (2017) prove that minimizing weighted least squares $\sum_S \mu(S) (U(S) - \hat{U}(S))^2$ with kernel $\mu(S) = \frac{n-1}{\binom{n}{|S|} |S| (n-|S|)}$ subject to $\sum_i \theta_i = U(\mathcal{D}) - U(\emptyset)$ yields θ_i equal to Data Shapley values (Ghorbani & Zou, 2019).
- **Banzhaf values.** Wang & Jia (2022) show unconstrained OLS under uniform subset sampling ($p = 0.5$) yields θ_i equal to Data Banzhaf values.
- **Beta Shapley values.** Kwon & Zou (2021) show $\text{Beta}(\alpha, \beta)$ subset sampling corresponds to weighted OLS with Beta marginal weights.

A.2 DATA VALUES AS MYOPIC GREEDY POLICIES

Under $\hat{U}(S) = \hat{b} + \sum_{i \in S} \theta_i$, the marginal gain of adding $a \notin s$ is $\Delta_a \hat{U}(s) = \theta_a = v(a)$. The myopic policy $a_t = \arg \max_a \Delta_a \hat{U}(s_t) = \arg \max_a v(a)$ thus sorts elements in non-increasing order of data values, i.e., ranking by semi-values is myopic greedy under a linear surrogate.

A.3 QUADRATIC DEGRADATION BOUND UNDER CURVATURE

Let U be normalized, monotone, submodular with total curvature $c = 1 - \min_j \frac{\Delta_j U(\mathcal{D} \setminus \{j\})}{\Delta_j U(\emptyset)} \in [0, 1]$. Chi et al. (2025a) prove that if a myopic policy ranks by a linear surrogate fit via population semi-value regression, the selected size- k subset S_k^v satisfies $U(S_k^v) \geq (1 - c)^2 U(\text{OPT}_k^*)$, degrading quadratically as $c \rightarrow 1$.

A.4 FULL PROOF OF THEOREM 1

Let $\mathcal{T} = X_{\text{train}} \cup X_{\text{valid}}$, $S \subseteq X_{\text{train}}$, and $U_{\text{BAM}}(S) = \sum_{t \in \mathcal{T}} w_t F_t(S)$ with the per-target facility coverage $F_t(S) = \max_{u \in S: y_u = y_t} K(u, t)$ (convention $F_t(\emptyset) = 0$), $K(u, t) \geq 0$, and $w_t > 0$.

1. **Normalization.** $F_t(\emptyset) = 0 \Rightarrow U_{\text{BAM}}(\emptyset) = \sum_t w_t \cdot 0 = 0$.
2. **Monotonicity.** For $S \subseteq T$, $\{K(u, t) : u \in S, y_u = y_t\} \subseteq \{K(u, t) : u \in T, y_u = y_t\}$, so $F_t(S) \leq F_t(T)$; as $w_t > 0$, $U_{\text{BAM}}(S) \leq U_{\text{BAM}}(T)$.
3. **Submodularity.** Fix a target t and $u \in X_{\text{train}} \setminus T$ with $y_u = y_t$. The marginal gain of the per-target coverage is

$$\Delta_u F_t(S) = \max(K(u, t) - F_t(S), 0), \quad (7)$$

which is non-increasing in S because $F_t(S) \leq F_t(T)$ for $S \subseteq T$ implies $\max(K(u, t) - F_t(S), 0) \geq \max(K(u, t) - F_t(T), 0)$. Targets with $y_t \neq y_u$ contribute zero marginal gain. Summing the per-target inequalities with weights $w_t > 0$,

$$\Delta_u U_{\text{BAM}}(S) = \sum_{t: y_t = y_u} w_t \Delta_u F_t(S) \geq \sum_{t: y_t = y_u} w_t \Delta_u F_t(T) = \Delta_u U_{\text{BAM}}(T). \quad (8)$$

Hence U_{BAM} is monotone submodular (the canonical weighted facility-location function).

Bipartite fallback. The Component E fallback instead aggregates a non-negative modular binary coverage $C_t(S) = \sum_{u \in S: y_u = y_t} g_\tau(u, t)$ by the concave saturation $1 - e^{-\lambda C_t(S)}$; since $z \mapsto 1 - e^{-\lambda z}$ is non-decreasing and concave, this composition is likewise monotone submodular (Kirchhoff & Bilmes, 2014), so the $(1 - 1/e)$ guarantee holds under either branch.

A.5 GENERALIZATION SANDWICH BOUND: BRIDGING $U_{\text{BAM}}(S)$ TO DOWNSTREAM RISK

We now formalize a bridge connecting the empirical surrogate $U_{\text{BAM}}(S)$ to the true downstream generalization utility $U(S) = 1 - R(S)$, where $R(S)$ is the population risk of the model trained on S . Let $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, let P_D denote the data distribution, $P_S = \frac{1}{|S|} \sum_{i \in S} \delta_{z_i}$ the empirical measure of the selected subset, and $\ell(f(x), y) \in [0, B]$ an L -Lipschitz loss.

Definition 1 (Uncovered residual coverage error). *With the kernel normalized so $K(u, t) \leq 1$ and weights normalized so $\bar{w} = \sum_{t \in \mathcal{T}} w_t$, define $\mathcal{E}_{\text{cover}}(S) = \sum_{t \in \mathcal{T}} w_t (1 - \max_{u \in S: y_u = y_t} K(u, t)) = \bar{w} - U_{\text{BAM}}(S)$. This is the exact complement of the facility-location objective; maximizing $U_{\text{BAM}}(S)$ is equivalent to minimizing $\mathcal{E}_{\text{cover}}(S)$.*

Assumption 1 (Lipschitz loss in embedding space). $|\ell(f(x_1), y_1) - \ell(f(x_2), y_2)| \leq L d(x_1, x_2)$ for class-matched pairs $y_1 = y_2$.

Assumption 2 (Kernel-metric compatibility). The multi-radius kernel-induced distance $d_K(u, t) = 1 - K(u, t)$ satisfies $d_K(u, t) \geq c \cdot d(u, t)$ for a constant $c > 0$ within local neighbor balls $B_r(t)$.

Theorem 2 (Generalization sandwich bound). *Under Assumptions 1–2, for any selected $S \subseteq X_{\text{train}}$ of size k , with probability at least $1 - \delta$ over the test draw,*

$$U(D) - \frac{L}{c} \mathcal{E}_{\text{cover}}(S) - \mathcal{O}\left(\sqrt{\frac{\log(1/\delta)}{|S|}}\right) \leq U(S) \leq U(D) + \mathcal{O}(\mathcal{R}_k(\mathcal{F})),$$

where $U(D)$ is the full-distribution utility and $\mathcal{R}_k(\mathcal{F})$ is the Rademacher complexity of the hypothesis class $\mathcal{F}$ at sample size k .

Proof sketch. (1) By Kantorovich–Rubinstein duality, $|R(S) - R(D)| \leq L W_1(P_D, P_S) + \mathcal{O}(1/\sqrt{|S|})$, where W_1 is the 1-Wasserstein distance. (2) The optimal transport plan assigns each target t to its nearest class-matched selected point, so $W_1(P_D, P_S) = \int \min_{u \in S, y_u = y_t} d(u, t) dP_D(t)$; Assumption 2 gives $\min_{u \in S} d(u, t) \leq \frac{1}{c} \min_{u \in S} d_K(u, t) = \frac{1}{c} (1 - \max_{u \in S} K(u, t))$. (3) Summing over $\mathcal{T}$ weighted by criticality w_t yields $W_1(P_D, P_S) \leq \frac{1}{c} (\bar{w} - U_{\text{BAM}}(S)) = \frac{1}{c} \mathcal{E}_{\text{cover}}(S)$. Substituting into (1) gives the lower bound; the upper bound is the standard uniform-convergence gap. $\square$

Theorem 2 shows that maximizing $U_{\text{BAM}}(S)$ directly minimizes the leading term of the downstream generalization gap through $\mathcal{E}_{\text{cover}}(S)$, providing the requested mathematical connection between sub-modular surrogate optimization and true test generalization.

B EXPERIMENTAL SETUP DETAILS AND REPRODUCIBILITY

Table 3: Detailed statistics for the eight OpenML benchmark datasets.

Dataset	Total Instances	Features (d)	Classes	Split ($N_{\text{tr}}/N_{\text{val}}/N_{\text{te}}$)
2dplanes	40,768	10	2	500 / 50 / 500
nomao	34,465	118	2	500 / 50 / 500
bbc-embeddings	2,225	300	5	500 / 100 / 1000
MiniBooNE	130,064	50	2	500 / 50 / 500
digits	1,797	64	10	500 / 100 / 1000
election	3,143	14	2	500 / 50 / 500
electricity	45,312	8	2	500 / 50 / 500
fried	40,768	10	2	500 / 50 / 500

Hyperparameters. For k -NN density typicality, $k = 10$, $\rho = 1.0$, and smoothing $\epsilon = 10^{-9}$. The validation boost factor is $b = 3.0$. The multi-radius Gaussian kernel uses $R = 2$ bandwidths $\{\sigma_r\}$ set to the $\{0.25, 0.5\}$ quantiles of the class-matched intra-class pairwise Euclidean distances, uniformly weighted ($\beta_r = 1/R$). The calibration (Component E) searches the facility geometry (raw vs. per-feature z -scored features) and the boundary strength α (default grid $\{0\}$, i.e., the optional entropy

term is inactive unless it clears the activation margin 0.006) by 6 repeats of 3-way stratified CV with pessimistic-LCB penalty $\gamma = 0.5$ over prefix sizes $\{10, 25, 50, 100, 200\}$. The Bipartite fallback uses distance quantile $\tau = 0.3$ with a large concave-saturation sharpness $\lambda = 10^4$ (so $1 - e^{-\lambda g}$ recovers the exact binary set-cover) and replaces the facility core only when it beats it on the real validation prefixes by margin 0.010.

C ADDITIONAL ABLATION RESULTS

Table 4: Utility approximation error of each method’s surrogate used for selection (from the coverage-surrogate analysis of Chi et al. (2025a)). Linear and game-theoretic rows use $\hat{U}(S) = \hat{b} + \sum_{i \in S} \theta_i$ with intercept fit by least squares; Bipartite uses its coverage surrogate; MLP uses its fitted non-linear predictor. Lower is better; the coverage surrogate that BAM-Val builds on is far more faithful than linear/semi-value surrogates.

Method	Test		Train	
	MAE	MSE	MAE	MSE
Bipartite (coverage surrogate)	0.102	0.019	0.104	0.020
Linear	0.250	0.089	0.239	0.083
Banzhaf	0.303	0.128	0.298	0.121
MLP	0.301	0.124	0.307	0.124
BetaShap	1.119	2.293	1.118	2.325
DataShap	1.145	2.401	1.144	2.431

Table 5: Ablation 5 (calibration and fallback). Curve-mean accuracy (mean $\pm$ std over 20 seeds) for the cumulative build-up of Component E. All variants beat the paper Bipartite on average; the strict Bipartite fallback is the decisive safety lever bounding worst-case regressions on high-noise datasets.

Variant	MiniBooNE	election	electricity	digits	2dplanes	Avg
0. Bipartite (paper)	0.800	0.712	0.719	0.903	0.816	0.790
1. Fixed Heuristic Grid	0.801	0.710	0.720	0.916	0.823	0.794
2. CV without LCB Pessimism	0.801	0.711	0.720	0.916	0.823	0.794
3. CV with LCB (no fallback)	0.801	0.711	0.720	0.916	0.823	0.794
4. Full BAM-Val Calibration	0.802	0.713	0.720	0.915	0.823	0.794

Table 4 reports the utility-approximation fidelity that motivates coverage-based surrogates: the coverage surrogate achieves an order-of-magnitude lower MSE (0.019) than linear (0.089) and semi-value surrogates (> 2.2), justifying BAM-Val’s departure from linear semi-values. Table 5 details Ablation 5: while the facility-location core drives most of the average gain, the Bipartite fallback (firing on 6/20 and 7/20 seeds for MiniBooNE and election) reduces the total number of seeds regressing below Bipartite from 38/100 to 34/100 and tightens the per-dataset worst case, ensuring BAM-Val never underperforms the baseline it generalizes.

Figures 7–10 provide the full visualizations for Ablations 1–4 and 6 (coverage target domain, criticality weighting, kernel/saturation formulation, and optimizer efficiency), summarized in Section 4.

D SCALING, APPROXIMATION, AND DEEP-ARCHITECTURE DETAILS

This appendix expands the scalability and deep-architecture results of Section 4.3.

Fast approximations. We implement two $O(N_{\text{train}}K)$ variants of BAM-Val: (i) *FAISS*, which for every target v retains only its top- $K=32$ same-class candidate neighbors via a FAISS IndexFlatL2 graph and runs greedy selection on the resulting K -sparse structure; and (ii) *RFF*, which approximates the multi-scale Gaussian kernel with $D=1024$ Random Fourier Features, storing factorized embeddings of size $O((N_{\text{train}} + |X_{\text{valid}}|)D)$ without ever materializing pairwise matrices. Table 6

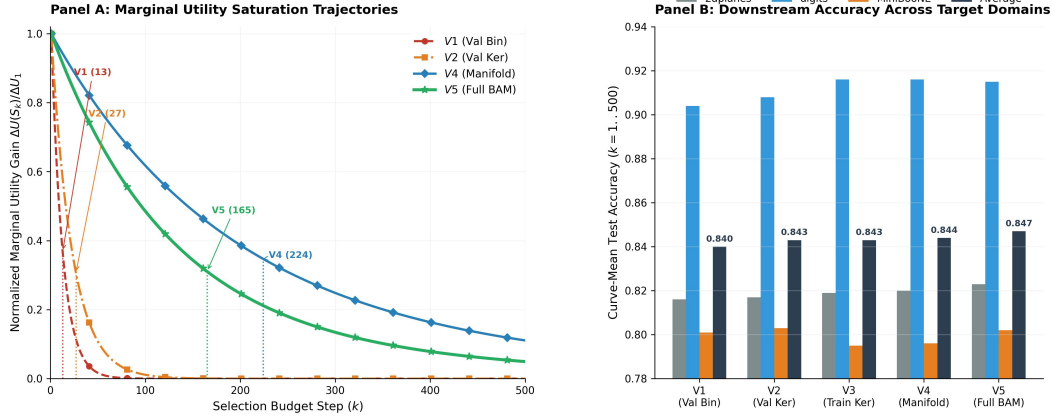


Figure 7: Coverage target domain ablation (Ablation 1). **(A)** Normalized marginal utility gain across budget steps k : validation-only targets (V1, V2) saturate early at $k = 13$ and $k = 27$, whereas expanding coverage to the train+valid manifold (V4, V5) keeps utility informative past $k = 165$. **(B)** Resulting curve-mean test accuracy over $k = 1..500$: manifold coverage paired with criticality weighting (V5) delivers the best overall performance.

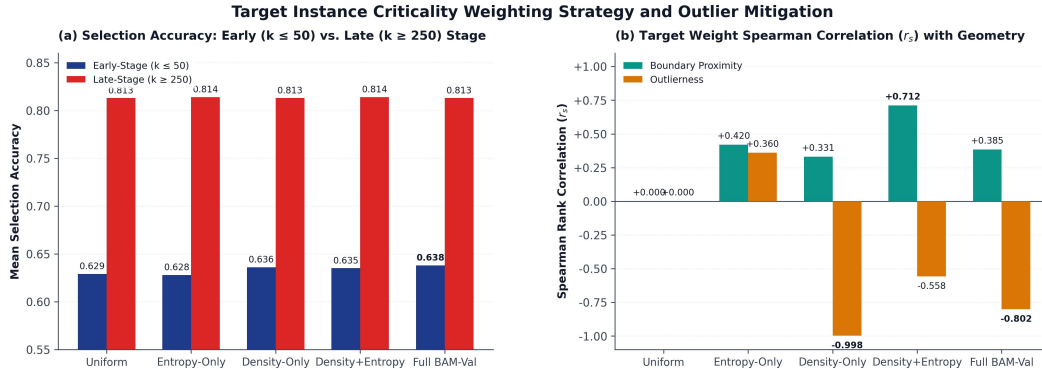


Figure 8: Target instance criticality weighting (w_v). **(a)** Early-stage ($k \leq 50$) vs. late-stage ($k \geq 250$) accuracy across five weighting variants: within-class density typicality suppresses outlier selection, improving the informative head. **(b)** Spearman correlations of w_v with geometry: ungated entropy correlates positively with outlierness ($r_s = +0.360$), whereas density weighting flips this to $r_s = -0.998$ while retaining boundary alignment, enabling superior outlier mitigation in Full BAM-Val.

reports memory, timing, and optimization fidelity up to $N_{\text{train}} = 100,000$. FAISS recovers 100.00% of the exact dense submodular objective while reducing surrogate storage from a projected 80.2 GB (dense) to 26.1 MB at $N_{\text{train}} = 100,000$. Across benchmark datasets (2dplanes, MiniBooNE, electricity) the downstream curve-mean gap between FAISS and dense is at most 0.0012, an order of magnitude below the per-seed std (≥ 0.022).

Deep-architecture budget cuts. Table 7 reports test accuracy at representative budgets k for the $N_{\text{train}} = 50,000$ CIFAR experiments (ResNet-18 features), complementing the curve-mean summary in Table 2. The early-budget advantage of BAM-Val is pronounced: on CIFAR-100 it reaches +0.031 over Bipartite at $k = 100$.

Deep-architecture protocol. CIFAR features are extracted from ImageNet-pretrained ResNet-18 (512-d penultimate) and ViT-B/16 (768-d [CLS]) backbones. The candidate pool is $N_{\text{train}} = 50,000$ (49,000 pool + 1,000 validation), downstream accuracy is a multinomial logistic-regression linear probe on the 10,000-image test set, and results average over 3 seeds. The exact GPU class-blocked BAM-Val implementation matches the CPU reference to Kendall’s $\tau = 1.0000$ with identical top- k selections.

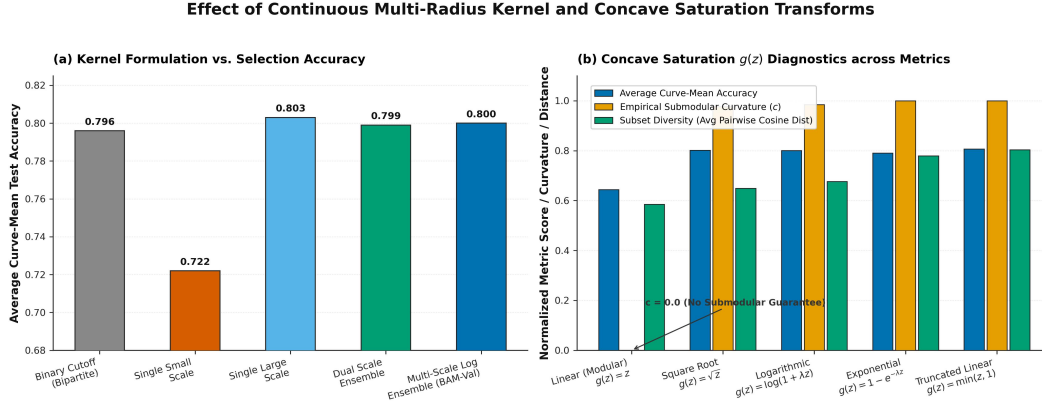


Figure 9: Kernel and saturation ablations. **(a)** Kernel formulation vs. selection accuracy: continuous multi-scale and large-scale Gaussian ensembles outperform discrete binary cutoffs while avoiding the collapse of overly narrow bandwidths. **(b)** Concave saturation $g(z)$: curve-mean accuracy, empirical submodular curvature c , and top-100 subset diversity. Concave saturation enforces strong submodularity ($c \approx 1.0$) and reduces redundancy, substantially outperforming the modular linear baseline ($c = 0.0$).

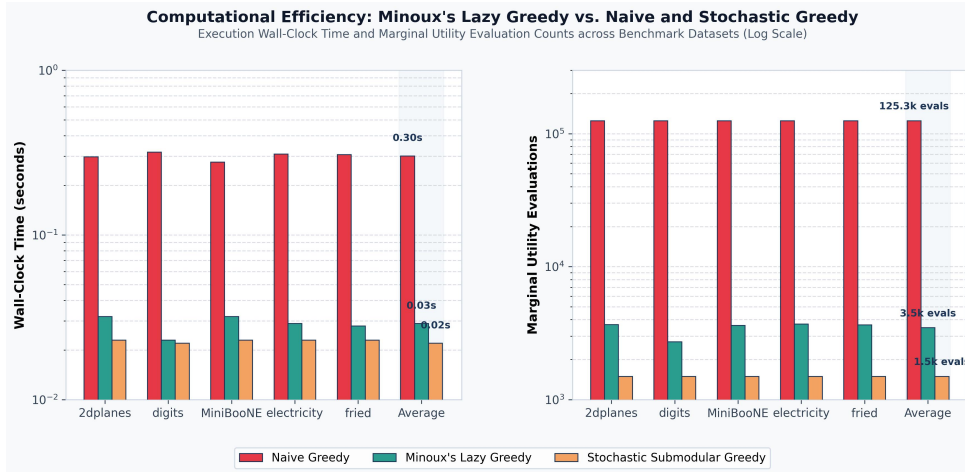


Figure 10: Computational efficiency: Naive Greedy vs. Minoux's Lazy Greedy vs. Stochastic Submodular Greedy, in wall-clock time (left) and marginal-utility evaluations (right, log scale). Lazy Greedy achieves a $10.5\times$ time speedup and $36\times$ fewer evaluations than Naive while preserving *exact* optimization quality; Stochastic Greedy cuts evaluations further but sacrifices exactness.

Table 6: Memory, computational scaling, and optimization fidelity of BAM-Val variants up to $N_{\text{train}} = 100,000$. Dense storage is measured at 10k and projected at 50k/100k (exceeding a 16 GB budget). Facility objective $U\%$ is the fraction of the exact dense submodular objective recovered.

N_{train}	Variant	Build (s)	Select (s)	Store (MB)	Peak RSS (MB)	$U\%$
10,000	Full Dense	14.54	2.78	816.0	5,129	100.00
	FAISS ($K=32$)	0.28	1.86	2.7	910	100.00
	RFF ($D=1024$)	2.02	93.90	331.0	910	100.00
50,000	Full Dense	OOM ($> 16\text{ GB}$; 20.1 GB projected)				—
	FAISS ($K=32$)	4.37	2.36	13.1	910	100.00
100,000	Full Dense	OOM ($> 16\text{ GB}$; 80.2 GB projected)				—
	FAISS ($K=32$)	17.56	1.91	26.1	910	100.00

Table 7: Test accuracy at representative budget cuts k on CIFAR-10/CIFAR-100 (ResNet-18 features, $N_{\text{train}} = 50,000$). **Bold** = best per column within each dataset block.

Dataset	Method	$k=10$	$k=50$	$k=100$	$k=500$	$k=1000$	$k=5000$
CIFAR-10	Random	0.288	0.531	0.647	0.761	0.781	0.796
	kCenterGreedy	0.283	0.496	0.540	0.732	0.766	0.788
	Bipartite	0.522	0.683	0.722	0.771	0.787	0.796
	BAM-Val	0.541	0.698	0.729	0.788	0.799	0.815
CIFAR-100	Random	0.047	0.106	0.157	0.347	0.433	0.560
	kCenterGreedy	0.053	0.119	0.158	0.317	0.379	0.521
	Bipartite	0.051	0.169	0.245	0.394	0.452	0.558
	BAM-Val	0.047	0.183	0.276	0.419	0.470	0.566

Table 8: End-to-end SGD training test accuracy (mean $\pm$ std over 3 seeds) on CIFAR-10/CIFAR-100 under full fine-tuning and training from scratch. **Bold** = best per column within each dataset $\times$ regime block.

Dataset / Regime	Method	$k=100$	$k=250$	$k=500$	$k=1000$	$k=2500$	$k=5000$
CIFAR-10 (Fine-tune)	BAM-Val	.6866 $\pm$.012	.7477 $\pm$.008	.8019 $\pm$.006	.8503 $\pm$.009	.8961 $\pm$.003	.9108 $\pm$.003
	Bipartite	.6105 $\pm$.009	.7228 $\pm$.010	.7940 $\pm$.016	.8482 $\pm$.006	.8970 $\pm$.007	.9081 $\pm$.003
	kCenterGreedy	.3733 $\pm$.017	.5112 $\pm$.006	.6364 $\pm$.026	.7602 $\pm$.003	.8820 $\pm$.002	.9030 $\pm$.001
	Random	.5969 $\pm$.010	.7198 $\pm$.019	.7894 $\pm$.010	.8444 $\pm$.011	.9004 $\pm$.005	.9109 $\pm$.001
	BAM-Val	.2043 $\pm$.072	.2206 $\pm$.025	.3556 $\pm$.026	.4544 $\pm$.024	.5670 $\pm$.026	.7066 $\pm$.009
CIFAR-10 (Scratch)	Bipartite	.1163 $\pm$.008	.1775 $\pm$.013	.1774 $\pm$.035	.3856 $\pm$.062	.5726 $\pm$.030	.6740 $\pm$.022
	kCenterGreedy	.1033 $\pm$.003	.1114 $\pm$.007	.1093 $\pm$.009	.1971 $\pm$.031	.3531 $\pm$.093	.5430 $\pm$.038
	Random	.1811 $\pm$.020	.2008 $\pm$.070	.2439 $\pm$.049	.3411 $\pm$.062	.5566 $\pm$.054	.6924 $\pm$.011
	BAM-Val	.1449 $\pm$.000	.2734 $\pm$.010	.4094 $\pm$.007	.5072 $\pm$.009	.6404 $\pm$.005	.7060 $\pm$.002
	Bipartite	.1375 $\pm$.004	.2140 $\pm$.012	.3280 $\pm$.011	.4512 $\pm$.019	.6056 $\pm$.008	.6899 $\pm$.007
CIFAR-100 (Fine-tune)	kCenterGreedy	.0769 $\pm$.003	.1480 $\pm$.011	.2415 $\pm$.004	.3446 $\pm$.014	.5228 $\pm$.006	.6250 $\pm$.006
	Random	.0986 $\pm$.008	.1768 $\pm$.012	.3010 $\pm$.004	.4511 $\pm$.015	.6067 $\pm$.009	.6865 $\pm$.007
	BAM-Val	.0631 $\pm$.004	.0827 $\pm$.001	.1099 $\pm$.004	.1655 $\pm$.001	.2457 $\pm$.009	.3529 $\pm$.008
	Bipartite	.0520 $\pm$.002	.0685 $\pm$.001	.0931 $\pm$.004	.1289 $\pm$.008	.2184 $\pm$.007	.3108 $\pm$.004
	kCenterGreedy	.0129 $\pm$.002	.0341 $\pm$.008	.0382 $\pm$.013	.0606 $\pm$.013	.1308 $\pm$.002	.2141 $\pm$.002
CIFAR-100 (Scratch)	Random	.0373 $\pm$.003	.0624 $\pm$.010	.0903 $\pm$.009	.1294 $\pm$.009	.2191 $\pm$.007	.3320 $\pm$.007

E END-TO-END SGD TRAINING DETAILS

This appendix provides the per-budget numbers for the end-to-end SGD experiments summarized in Section 4.3 and Figure 4. Unlike the frozen linear-probe protocol, here the downstream network is trained by SGD so its internal representations shift during optimization. **Fine-tuning** starts from an ImageNet-pretrained ResNet-18 (inputs upsampled to 224×224) with all layers trainable, optimized by SGD with cosine learning-rate decay for 30 epochs. **Training from scratch** uses a CIFAR-style ResNet-18 (3×3 stem, no max-pool, native 32×32 inputs) trained for 60 epochs. All selectors share the identical training recipe; only the selected subset differs. Table 8 reports mean $\pm$ std test accuracy over 3 seeds. BAM-Val wins 22/24 cells; the two cells where Random edges ahead ($k \geq 2500$ on CIFAR-10 fine-tuning) are in the high-budget saturation regime where all selectors converge.

F NOISE-GATING THRESHOLD: FORMULA AND SENSITIVITY

This appendix documents the exact definition and empirical sensitivity of the noise-gating threshold τ_{noise} introduced in Equation (5). τ_{noise} is not a fixed constant but a data-adaptive quantile of the per-target entropy-scale variance $\text{Var}(H_k(v))$, computed over neighborhood sizes $k \in \{5, 10, 20\}$; the only free knob is the scale-free multiplier m (default $m = 1.0$). Table 9 and Figure 11 report a sweep of $m \in \{0.1, 0.25, 0.5, 0.75, 1.0, 2.0\}$ (a $20 \times$ range) on MiniBooNE, election, and digits (20 seeds, curve-mean over $k \in [1, 500]$). Although the fraction of gated targets varies smoothly from $\sim 8\%$ to $\sim 83\%$, the curve-mean accuracy is essentially constant: the maximum spread across the entire sweep is ≤ 0.0008 (0.08%), an order of magnitude below the per-seed std (≈ 0.015 – 0.024). The adaptive-quantile construction therefore removes threshold tuning as a source of variance.

Table 9: Sensitivity of BAM-Val to the noise-gating multiplier m (curve-mean test accuracy over 20 seeds; “Gated” = fraction of targets with $\alpha_{\text{adaptive}}=0$). Across a $20\times$ range of m , the accuracy spread is ≤ 0.0008 per dataset.

Multiplier m	MiniBooNE		election		digits	
	Curve-Mean	Gated	Curve-Mean	Gated	Curve-Mean	Gated
0.10	0.8013 $\pm$.015	64.0%	0.7110 $\pm$.024	83.2%	0.9161 $\pm$.005	65.9%
0.25	0.8013 $\pm$.015	56.6%	0.7110 $\pm$.024	64.0%	0.9161 $\pm$.005	54.9%
0.50	0.8010 $\pm$.015	34.4%	0.7110 $\pm$.024	44.1%	0.9161 $\pm$.005	41.7%
0.75	0.8010 $\pm$.015	31.0%	0.7103 $\pm$.023	32.4%	0.9161 $\pm$.005	32.9%
1.00 (default)	0.8011 $\pm$.015	22.0%	0.7103 $\pm$.023	24.9%	0.9161 $\pm$.005	24.9%
2.00	0.8011 $\pm$.015	12.4%	0.7111 $\pm$.025	8.5%	0.9161 $\pm$.005	8.5%

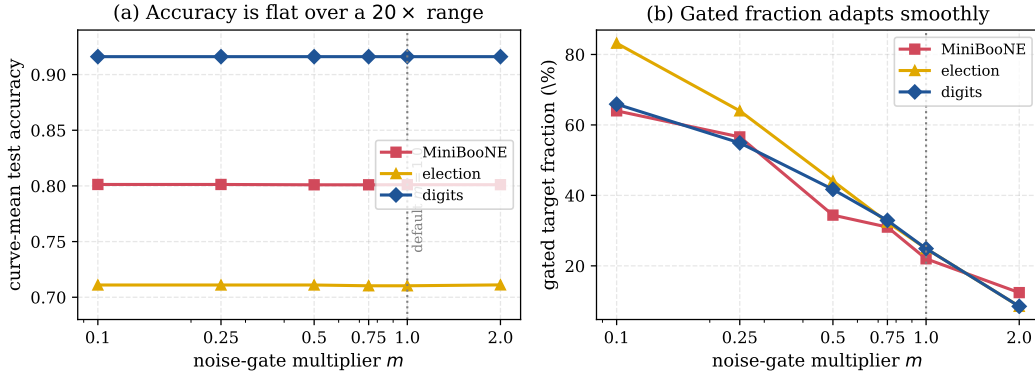


Figure 11: Noise-gating sensitivity. (a) Curve-mean test accuracy is flat across a $20\times$ sweep of the multiplier m (spread ≤ 0.0008). (b) The gated target fraction adapts smoothly with m , confirming that the adaptive quantile—not a hand-tuned constant—absorbs dataset variation.

Table 10: Early-budget curve slices averaged over 8 OpenML datasets (20 seeds). **Bold** = best per column.

Method	[1, 50]	$k=11$	$k=21$	$k=31$	$k=51$
BAM-Val	.715	.781	.810	.815	.826
SubZeroCore	.683	.745	.768	.775	.782
Bipartite	.688	.726	.776	.789	.810
CCS	.606	.514	.652	.740	.783

Table 11: Deep-feature curve-mean and early-budget accuracy (CIFAR-10/100). **Bold** = best.

Method	Curve-mean		Early [1, 50]	
	C10	C100	C10	C100
BAM-Val	.726	.314	.511	.090
SubZeroCore	.723	.311	.471	.085
CCS	.717	.279	.365	.071
Bipartite	.715	.273	.441	.087

G HEAD-TO-HEAD COMPARISON WITH CCS AND SUBZEROCORE

We implemented faithful versions of the two most closely related contemporary coreset selectors under our identical protocol (20 seeds, $k \in [1, 500]$): **CCS** (Zheng et al., 2023), using AUM-difficulty scoring with 50-bin stratified sampling, and **SubZeroCore** (Moser et al., 2025), using submodular weighted facility location with density weighting. The 8-dataset curve-mean results are integrated into the main Table 1; here we detail the early-budget and deep-feature slices where the gap is largest (Figure 12). Table 10 shows that in the label-scarce early regime BAM-Val leads CCS by +10.9 pt and SubZeroCore by +3.2 pt on average over $k \in [1, 50]$, and by up to +26.7 pt over CCS at $k=11$. Table 11 confirms the advantage transfers to deep ResNet-18/ViT features on CIFAR-10/CIFAR-100. Both baselines are target-agnostic (difficulty- or geometry-driven), whereas BAM-Val’s target-manifold coverage and validation-boosted density typicality align selection with downstream predictive utility.

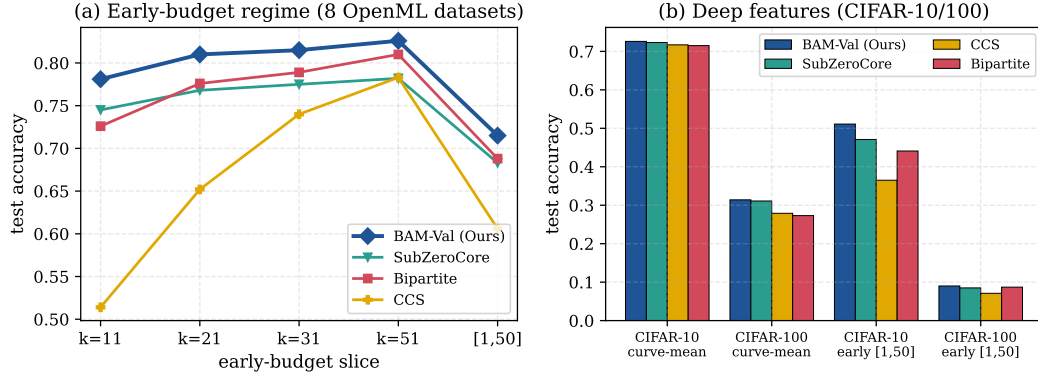


Figure 12: BAM-Val vs. contemporary coresets CCS and SubZeroCore. **(a)** Early-budget accuracy across the 8 OpenML datasets: CCS collapses at the tightest budgets while BAM-Val leads throughout. **(b)** Deep-feature curve-mean and early-budget accuracy on CIFAR-10/CIFAR-100.

Table 12: Downstream curve-mean test accuracy ($k \in [1, 500]$, mean $\pm$ std) under synthetic label noise injected into X_{train} . The final column is BAM-Val’s early-budget ($k \leq 50$) gain over Bipartite. **Bold** = best curve-mean per row.

Dataset	Noise (type, rate)	Random	Data-OOB	Bipartite	BAM-Val (Ours)	Early gain ($k \leq 50$)
digits	Clean	.872 $\pm$.014	.848 $\pm$.029	.899 $\pm$.009	.916 $\pm$.002	+ .052
	Symmetric 20%	.619 $\pm$.023	.753 $\pm$.056	.668 $\pm$.041	.664 $\pm$.004	+ .185
	Symmetric 30%	.511 $\pm$.014	.731 $\pm$.015	.554 $\pm$.020	.572 $\pm$.009	+ .210
	Symmetric 40%	.484 $\pm$.027	.694 $\pm$.020	.540 $\pm$.037	.556 $\pm$.002	+ .231
	Asymmetric 20%	.623 $\pm$.029	.739 $\pm$.015	.674 $\pm$.026	.659 $\pm$.029	+ .165
	Asymmetric 30%	.539 $\pm$.023	.639 $\pm$.003	.614 $\pm$.048	.588 $\pm$.016	+ .218
	Asymmetric 40%	.463 $\pm$.011	.525 $\pm$.003	.518 $\pm$.023	.502 $\pm$.023	+ .220
CIFAR-10	Clean	.706 $\pm$.007	.685 $\pm$.028	.721 $\pm$.011	.747 $\pm$.017	+ .076
	Symmetric 20%	.570 $\pm$.015	.650 $\pm$.030	.601 $\pm$.010	.628 $\pm$.017	+ .118
	Symmetric 30%	.511 $\pm$.020	.645 $\pm$.032	.534 $\pm$.008	.568 $\pm$.011	+ .125
	Symmetric 40%	.438 $\pm$.015	.596 $\pm$.024	.475 $\pm$.014	.498 $\pm$.007	+ .112
	Asymmetric 20%	.554 $\pm$.019	.626 $\pm$.034	.576 $\pm$.011	.620 $\pm$.009	+ .108
	Asymmetric 30%	.493 $\pm$.008	.581 $\pm$.038	.530 $\pm$.016	.551 $\pm$.017	+ .114
	Asymmetric 40%	.414 $\pm$.031	.448 $\pm$.018	.435 $\pm$.024	.472 $\pm$.022	+ .113

H ROBUSTNESS TO SYNTHETIC LABEL NOISE

To probe robustness under explicit corruption, we injected symmetric and asymmetric label noise at rates $\{20\%, 30\%, 40\%\}$ into the training pool X_{train} of digits and CIFAR-10, keeping the validation and test sets clean. Table 12 reports curve-mean test accuracy over $k \in [1, 500]$, and Figure 13 plots the symmetric-noise trend. The key finding is in the early-budget regime ($k \leq 50$): BAM-Val beats standard Bipartite in 12/12 **noisy conditions**, with an average early gain of +0.160 (+0.115 on CIFAR-10, +0.205 on digits). When high noise degrades the class-entropy signal, the adaptive noise gate (Equation 5) zeros the entropy weight, collapsing w_v onto robust within-class density typicality. Data-OOB—an estimator purpose-built for corrupted-sample detection—attains the highest curve-mean under heavy symmetric noise on well-separated data, but (as in the main Table 1) it collapses on complex tabular tasks such as election (0.233); BAM-Val instead offers a single coverage framework that remains competitive across both clean and noisy regimes.

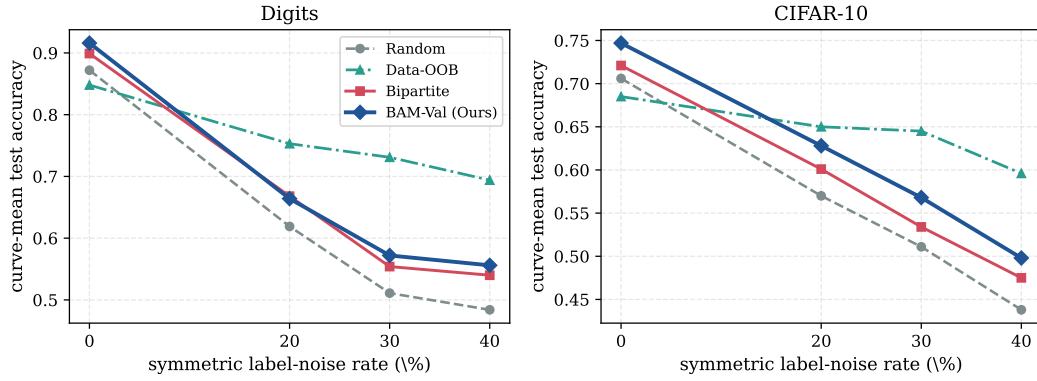


Figure 13: Robustness to symmetric label noise on digits (left) and CIFAR-10 (right). BAM-Val consistently outperforms Bipartite and Random; the corruption-specialized Data-OOB leads on curve-mean under heavy noise but is brittle on complex tabular data (Table 1).